■ FEATURES OF 1VD-FTV ENGINE

The 1VD-FTV engine has been able to achieve the following performance through the adoption of the items listed below.

(1) High performance and reliability
(2) Low noise and vibration
(3) Lightweight and compact design
(4) Good serviceability
(5) Clean emission and fuel economy

Item		(1)	(2)	(3)	(4)	(5)
Engine Proper	Cylinder head cover is made of plastic and cylinder head is made of aluminum alloy.	○		○		
	A cylinder block made of compact graphite iron is used.		○	○		
	Piston provided in combustion chamber is used in conjunction with the adoption of direct injection.	○				○
	The piston skirt is coated with resin.	○				○
	The passage for the EGR system is provided in the cylinder head. (Only on models with EGR)			○		○
	Crankshaft pulley uses an inertial weight.		○			
	A No.1 oil pan sub-assembly made of aluminum alloy is used.		○			
Valve Mechanism	A camshaft is driven by a timing chain and gear.	○	○	○		
	Hydraulic lash adjusters are used.	○	○		○	
	Roller rocker arms are used.	○	○			○
Lubrication System	An oil filter with a replaceable element is used.				○	
	A water-cooled type engine oil cooler is used.	○				
	A scavenging pump is used.	○				
Cooling System	TOYOTA Genuine SLLC (Super Long Life Coolant) is used as the engine coolant.				○	
Intake and Exhaust System	The rotary solenoid type diesel throttle motor and a non-contact throttle position sensor are used in the diesel throttle body.					○
	A variable nozzle vane type turbocharger driven by a DC motor is used in each bank.	○		○		○
	An air-cooled type intercooler is used.	○				○
	The linear solenoid type EGR valve is used. (Only on models with EGR)					○
	A water-cooled type EGR cooler is used. (Only on models with EGR)					○
	Oxidation catalytic converter is used. (Only on models complying with the emission regulations*)					○

(Continued)

Item		(1)	(2)	(3)	(4)	(5)
Fuel System	3-plunger type fuel supply pump is used.	○	○			
	A common-rail type fuel injection system is used.	○	○			○
	Solenoid type fuel injectors on which compensation value and QR (Quick Response) code are printed are used.	○	○			○
	A fuel filter, in which the fuel filter element alone can be replaced, is used.				○	
	A fuel filter warning switch that detects the clogging of the fuel filter is used.	○			○	
	Water-cooled and air-cooled type fuel coolers are used.	○				○
Charging System	Segment conductor type generator is used.			○		
Engine Mount	The electrical hydraulic type engine mountings are used.		○			
Serpentine Belt Drive System	A serpentine belt drive system is used.			○	○	
Engine Control System	A pilot injection control is used.	○	○			○
	A fuel pump control, which has the fuel transfer function, is used. (Only on models with dual fuel tank)					○
	The cranking hold function is used.	○				
	An oil maintenance management system is used. (Only on European models complying with EURO IV)				○	

*: Except automatic transmission models complying with EURO II